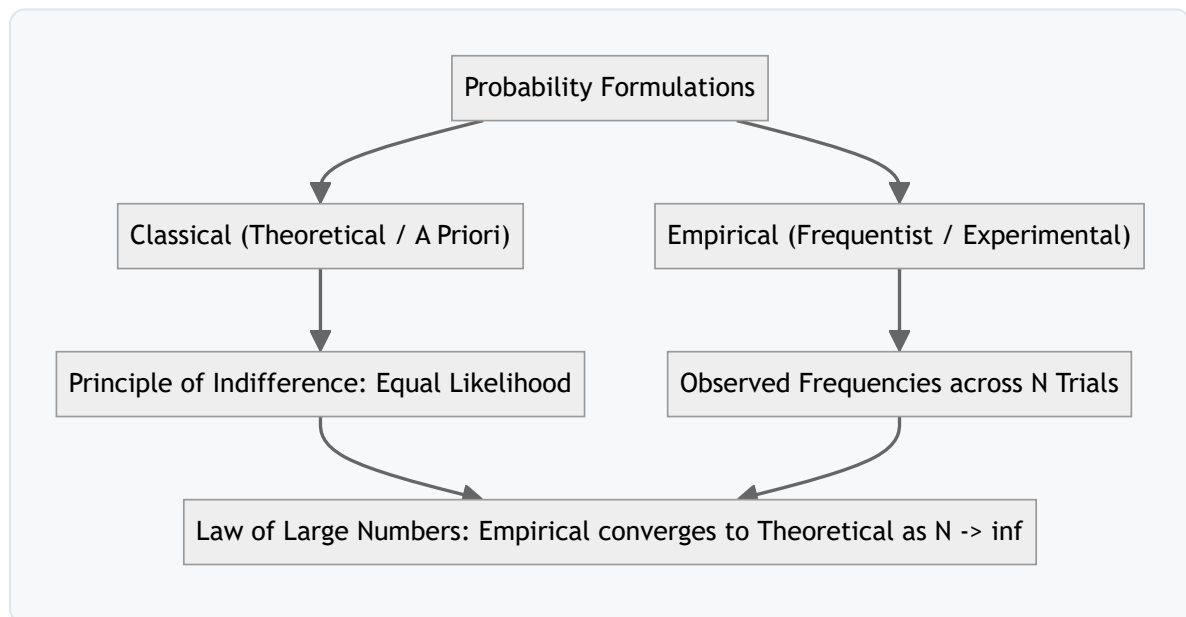


Day 20 - Lecture 20.3: Classical vs. Empirical Probability

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English](#) | [हिंदी / Hinglish](#) | [Back to Lecture 20 Module](#)

1. Two Philosophical Interpretations of Probability

In statistics and machine learning, probability is quantified through two complementary frameworks: **Classical (A Priori)** and **Empirical (Frequentist)**.



Comparison Matrix:

Dimension	Classical Probability ($P_{extclass}$)	Empirical Probability (P_{extemp})
Origin	Theoretical deduction prior to experiment	Observation of actual experiment outcomes
Assumption	All elementary outcomes are equally likely	No assumption of symmetry; learned from data
Formula	$P(A) = \frac{n(A)}{n(\mathcal{S})}$	$P(A) = \frac{\text{Count of } A}{N \text{ total trials}}$
Domain	Games of chance (Dice, Cards, Coins)	Machine Learning training datasets

2. Mathematical Formalism

Classical (Laplacian) Probability:

For a finite sample space $\mathcal{S}$ with $|\mathcal{S}|$ equally likely outcomes:

$$P(A) = \frac{|A|}{|\mathcal{S}|} = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Empirical (Frequentist) Probability:

In an experiment repeated independently N times under identical conditions, let $n_N(A)$ denote the frequency of event A :

$$\hat{P}_N(A) = \frac{n_N(A)}{N}$$

The Law of Large Numbers (Convergence Theorem):

As the number of experimental repetitions N approaches infinity, the empirical relative frequency converges almost surely to the true theoretical probability $P(A)$:

$$\lim_{N \rightarrow \infty} P\left(\left|\frac{n_N(A)}{N} - P(A)\right| \geq \epsilon\right) = 0 \quad \text{for all } \epsilon > 0$$

3. Python Simulation: Law of Large Numbers

```
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
n_flips = 10000

# Simulate coin tosses where 1 = Heads, 0 = Tails (Fair coin: P(H) = 0.5)
tosses = np.random.binomial(1, 0.5, size=n_flips)
cumulative_heads = np.cumsum(tosses)
trial_numbers = np.arange(1, n_flips + 1)
empirical_probabilities = cumulative_heads / trial_numbers

print(f"After 10 flips:    P_hat(H) = {empirical_probabilities[9]:.4f}")
print(f"After 100 flips:   P_hat(H) = {empirical_probabilities[99]:.4f}")
print(f"After 1,000 flips: P_hat(H) = {empirical_probabilities[999]:.4f}")
print(f"After 10,000 flips: P_hat(H) = {empirical_probabilities[-1]:.4f}")
print(f"Theoretical Target: P(H)    = 0.5000")
```

4. Key Takeaways & Interview Points

- **Machine Learning is Empirical:** Machine learning algorithms estimate probability distributions from finite training data (P_{temp}), not theoretical proofs.
- **Law of Large Numbers (LLN):** With larger datasets, empirical estimates stabilize toward the true underlying data-generating distribution.